

- 5 Fig. 5.1 shows the arrangement of the apparatus used in the measurement of the magnetic flux density between the poles of a U-shaped magnet. The magnet rests on top of an electronic balance. A stiff rectangular frame measuring 8.0 cm by 5.0 cm carries a current I between the magnetic poles. The bottom edge of the frame is lowered into the region between the poles of the U-shaped magnet and is entirely within the uniform magnetic field.

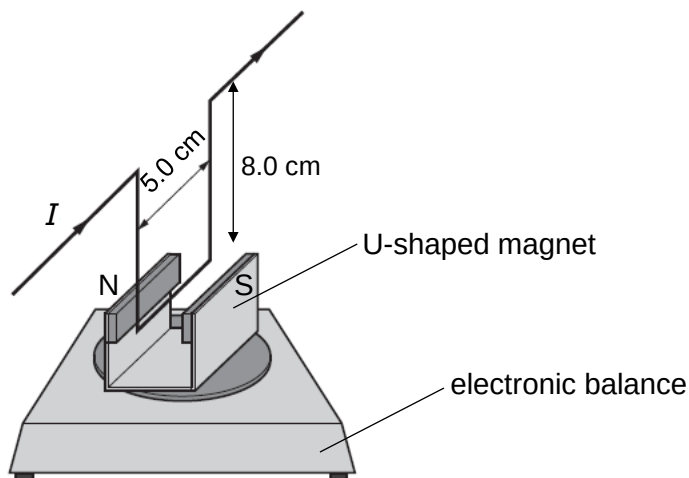


Fig. 5.1

Fig. 5.2 shows the apparatus when viewed from the front. The poles of the U-shaped magnet have been indicated.

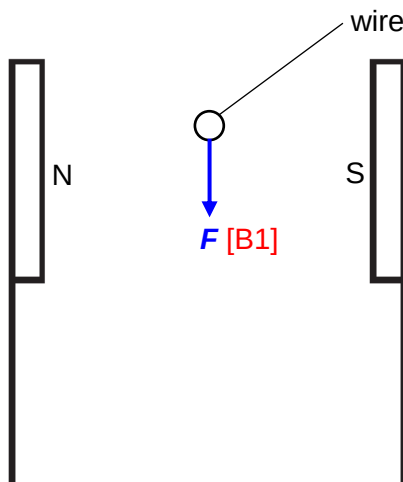


Fig. 5.2

- (a) (i) Draw, on Fig. 5.2, a vertical force acting on the wire when current passes through it. Label the force F . [1]
- (ii) When current passes through the wire, the balance reading changes.

Explain why there is a change in reading, and whether the change is an increase or decrease.

By Newton's 3rd Law, the interaction between wire and magnet causes an upward force on magnet [M1] which causes balance reading to decrease. [A1 – allow e.c.f. from (a)(i)] [2]

- (b) The current I is varied and the change in the balance reading is recorded as shown in Fig. 5.3.

I / A	Change in balance reading / g			Mean change / g	$F / \times 10^{-3} \text{ N}$
	Trial 1	Trial 2	Trial 3		
0.5	0.08	0.05	0.06	0.06	0.6
1.0	0.14	0.16	0.16	0.15	1.5
1.5	0.22	0.20	0.23	0.22	2.2
2.0	0.31	0.29	0.31	0.30	2.9
2.5	0.38	0.39	0.34	0.37	3.6
3.0	0.44	0.48	0.48	0.47	4.6

Fig. 5.3

- (i) Complete the table in Fig. 5.3. [1]
- (ii) On Fig. 5.4, plot the missing data point from the table in Fig. 5.3 and draw a line of best fit. [1]

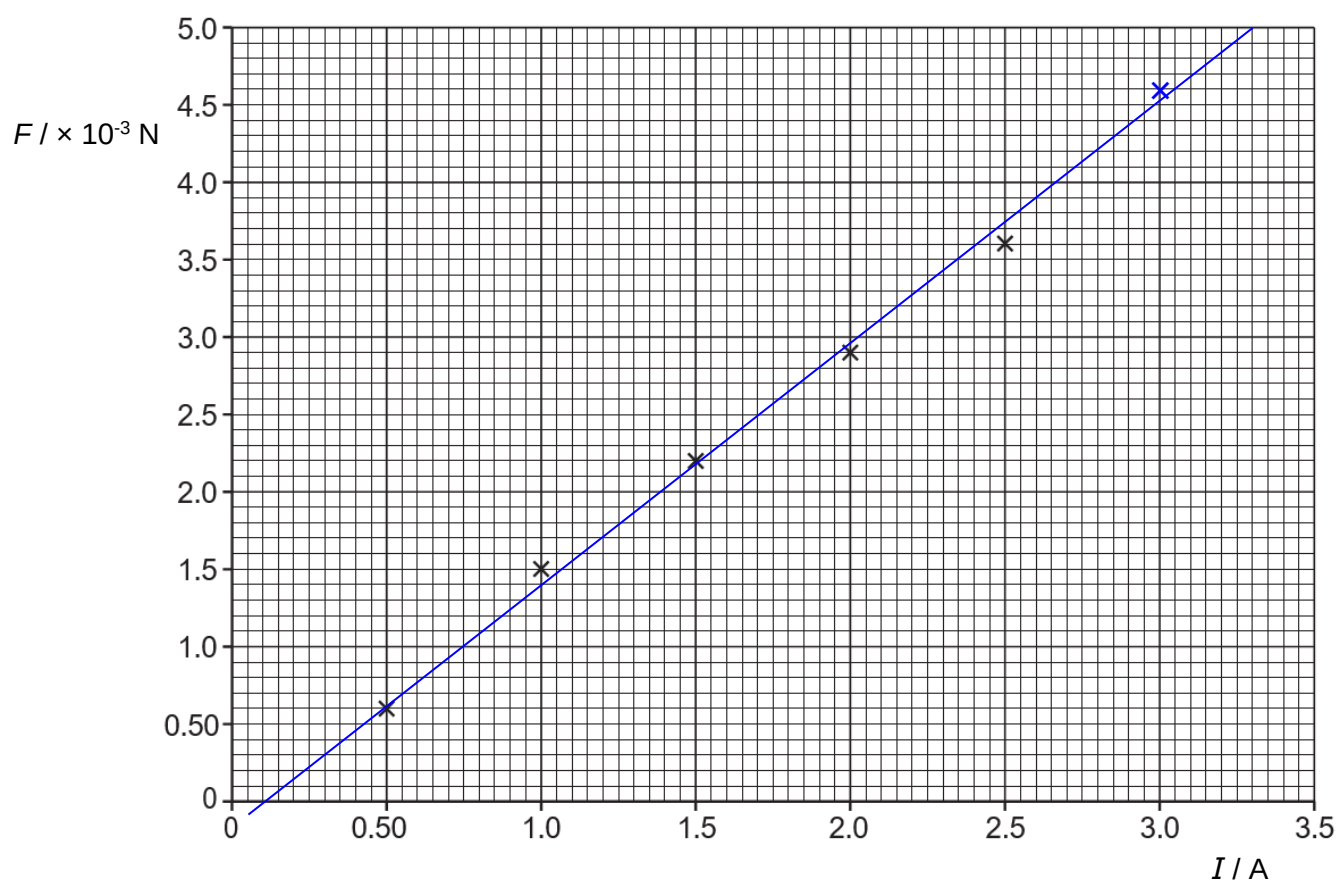


Fig. 5.4

- (iii) Use your graph to determine the value of the magnetic flux density B , in mT, between the U-shaped magnet.

$$\text{gradient} = \frac{5.0 \times 10^{-3} - 0}{3.3 - 0.10}$$

$$= 0.00156 \quad [\text{M1 - ignore POT error}]$$

$$BL = 0.00156$$

$$B = \frac{0.00156}{0.050}$$

$$= 0.0312 \text{ T}$$

$$= 31 \text{ mT (acceptable range: 28 mT to 34 mT)} \quad [\text{A1}]$$

$$B = \dots\dots\dots \text{ mT} \quad [2]$$

- (iv) The value of the magnetic flux density in (b)(iii) is slightly inaccurate for a reason which has nothing to do with human error or meter inaccuracies.

Suggest one reason that might be the cause of this inaccuracy.

Effect of the Earth's magnetic field / Presence of external magnetic field [B1]

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..... [1]

[Total: 8]